

# Data Analysis

- Measures of Central tendency or average
- Measures of Variation or dispersion
- Measures of Skewness
- Measures of Kurtosis

# Measures of Central tendency or average

- ☐ Arithmetic Mean
- ☐ Median
- ☐ Mode
- ☐ Geometric Mean
- ☐ Harmonic Mean

## Arithmetic Mean (A.M.)

**A.M.** : Arithmetic mean of set of observations is their sum divided by the number of observations.

$$\text{Simple A.M. : } \bar{X} = \frac{x_1 + x_2 + \dots + x_n}{n} = \frac{\sum_{i=1}^n x_i}{n}$$

$n$  - Number of observations.

**Ex:** A Monthly income of 10 families in a city is given by:

1600, 1560, 1440, 1530, 1670, 1860, 1750, 1910, 1490, 1800.

$$\bar{X} = \frac{\sum_{i=1}^{10} x_i}{10} = \frac{16610}{10} = \mathbf{1661}$$

## A.M. For Discrete Series:

Direct Method:  $\bar{X} = \frac{\sum_{i=1}^n f_i x_i}{n}$

$f$  - Frequency of the given set of observations

$n = \sum_{i=1}^n f_i$  = Total number of observations

**Ex:** The following data represents the marks obtained by 60 students of a class. Obtain the average marks.

| Marks           | 20 | 30 | 40 | 50 | 60 | 70 |
|-----------------|----|----|----|----|----|----|
| No. of Students | 8  | 12 | 20 | 10 | 6  | 4  |

| <b>Marks</b> | <b>No. of Students</b> | <b><math>fx</math></b> |
|--------------|------------------------|------------------------|
| 20           | 8                      | 160                    |
| 30           | 12                     | 360                    |
| 40           | 20                     | 800                    |
| 50           | 10                     | 500                    |
| 60           | 6                      | 360                    |
| 70           | 4                      | 280                    |
|              | $n = 60$               | $\Sigma fx = 2460$     |

$$\bar{X} = \frac{\Sigma f_i x_i}{n} = \frac{2460}{60} = \mathbf{41}$$

## A.M. For Continuous Series:

Direct Method:  $\bar{X} = \frac{\sum_{i=1}^n f_i x_i}{n}$

$f$  - Frequency of the given set of observations

$x$  - mid-point each class

$n = \sum_{i=1}^n f_i$  = Total number of observations

**Ex:** Obtain the Arithmetic mean for the following data.

| Marks           | 0-10 | 10-20 | 20-30 | 30-40 | 40-50 | 50-60 |
|-----------------|------|-------|-------|-------|-------|-------|
| No. of Students | 12   | 18    | 27    | 20    | 17    | 6     |

| <b>Marks</b> | <b>No. of Students<br/>(<math>f</math>)</b> | <b>Mid-point<br/>(<math>x</math>)</b> | <b><math>fx</math></b> |
|--------------|---------------------------------------------|---------------------------------------|------------------------|
| 0-10         | 12                                          | 5                                     | 60                     |
| 10-20        | 18                                          | 15                                    | 270                    |
| 20-30        | 27                                          | 25                                    | 675                    |
| 30-40        | 20                                          | 35                                    | 700                    |
| 40-50        | 17                                          | 45                                    | 765                    |
| 50-60        | 6                                           | 55                                    | 330                    |
|              | $n = \sum f_i = 100$                        |                                       | $\sum fx = 2800$       |

$$\bar{X} = \frac{\sum f_i x_i}{n} = \frac{2800}{100} = 28$$

# Using the Deviation:

If the values of  $x$  or  $f$  are large, the calculation of A.M. by above formula is quite time-consuming and tedious.

The A.M. is reduced to a great extent by taking the deviations of the given values from any arbitrary point 'A':

$$\text{Let } d_i = x_i - A \Rightarrow f_i d_i = f_i (x_i - A)$$

$$\bar{X} = A + \frac{1}{n} \sum_{i=1}^n f_i d_i \quad \text{or}$$

$$\bar{X} = A + \frac{h}{n} \sum_{i=1}^n f_i d_i$$

*h – common magnitude of  
class interval*



**Ex:**

| C.I.      | 0-8 | 8-16 | 16-24 | 24-32 | 32-40 | 40-48 |
|-----------|-----|------|-------|-------|-------|-------|
| Frequency | 8   | 7    | 16    | 24    | 15    | 7     |

| C.I.  | Mid-Value | Frequency | $d = \frac{x - A}{h}$ | $fd$ |
|-------|-----------|-----------|-----------------------|------|
| 0-8   | 4         | 8         | -3                    | -24  |
| 8-16  | 12        | 7         | -2                    | -14  |
| 16-24 | 20        | 16        | -1                    | -16  |
| 24-32 | 28        | 24        | 0                     | 0    |
| 32-40 | 36        | 15        | 1                     | 15   |
| 40-48 | 44        | 7         | 2                     | 14   |
| Total |           | 77        |                       | -25  |

$$\bar{X} = A + \frac{h}{n} \sum_{i=1}^n f_i d_i = 28 + \frac{8 \times (-25)}{77} = 25.404$$

# Mathematical Properties of A.M.

1. The sum of the deviations of the observations from the arithmetic mean is always zero. i.e.,  $\sum(X - \bar{X}) = 0$
2. The sum of the squared deviations of the items from the arithmetic mean is minimum, i.e., less than the sum of the squared deviations of the observations from any other value.  $\sum(X - \bar{X})^2 = 0$ .
3. If  $\bar{X}_1$  and  $\bar{X}_2$  are the A.M. of two related groups, then  $\bar{X}_{12}$  – is the combined average of the these two groups :

$$\bar{X}_{12} = \frac{N_1\bar{X}_1 + N_2\bar{X}_2}{N_1 + N_2}$$

# Weighted Arithmetic Mean

When the relative importance of the different observations is not the same, we need to compute the Weighted A.M. ( $\bar{X}_w$ ) :

$$\bar{X}_w = \frac{\sum W_i X_i}{\sum W_i}$$

# Example

A contractor employs three categories of workers. He pays Rs. 30 per day to Category A workers, Rs. 20 per day to Category B workers, and Rs. 10 per day to Category C workers. What is the average wage per day paid by the contractor?

# Example

A train runs 25 kilometers at a speed of 30 km/hr, another 50 km at speed of 40 km/hr., then due to repairs on the track, the train travels for 6 minutes at a speed of 10 km/hr and finally covers the remaining distance of 24 kilometers at a speed of 24 km/hr.

What is the average speed in kilometers per hour ?

| <b>Speed (<math>X</math>)<br/>(km/hr)</b> | <b>Time (<math>W</math>)<br/>(minutes)</b> | <b>Weights<br/>(<math>WX</math>)</b> |
|-------------------------------------------|--------------------------------------------|--------------------------------------|
| 30                                        | 50                                         | 1500                                 |
| 40                                        | 75                                         | 3000                                 |
| 10                                        | 6                                          | 60                                   |
| 24                                        | 60                                         | 1440                                 |
|                                           | $\Sigma W = 191$                           | $\Sigma WX = 6000$                   |

$$\bar{X} = \frac{\sum X_i W_i}{\sum W_i} = \frac{6000}{191} = \mathbf{31.41 \text{ km/hr}}$$

# Median

Median is the value of the variable which divides the entire distribution into two equal parts.

It is the value which exceeds and is exceeded by the same number of observations. Thus the median is called as a “*positional average*”.

*Evaluation of Median:* For ungrouped data,

- (i) Odd number of observations.
- (ii) Even number of observations.

***For Un-Grouped data:***

Arrange the data in ascending or descending order of magnitude.

a) Find  $\frac{N+1}{2}$ , where **N** – Total number of observations.

b) Median = Size of  $\frac{N+1}{2}$  *th* item in the series.



# Example

Obtain the median from the following data:

391, 384, 407, 672, 522, 777, 753, 2488, 1490

# Example

Obtain the median from the following data:

891, 884, 991, 1072, 922, 1277, 1153, 2488, 1490

***For Grouped data:***

(i) Discrete Frequency distribution:

a) Find  $\frac{N+1}{2}$ , where  $N$  – Total Frequency =  $\sum_{i=1}^N f_i$ .

b) See the cumulative frequency (c.f.) value, equal or just greater than the  $\frac{N+1}{2}$ .

c) The corresponding value of  $x$  is median.

# Example

Obtain the median from the following data:

|     |   |    |    |    |    |    |    |   |   |
|-----|---|----|----|----|----|----|----|---|---|
| $x$ | 1 | 2  | 3  | 4  | 5  | 6  | 7  | 8 | 9 |
| $f$ | 8 | 10 | 11 | 16 | 20 | 25 | 15 | 9 | 6 |

| $x$ | $f$     | $c.f.$ |
|-----|---------|--------|
| 1   | 8       | 8      |
| 2   | 10      | 18     |
| 3   | 11      | 29     |
| 4   | 16      | 45     |
| 5   | 20      | 65     |
| 6   | 25      | 90     |
| 7   | 15      | 105    |
| 8   | 9       | 114    |
| 9   | 6       | 120    |
|     | N = 120 |        |

$$\frac{N + 1}{2} = 60.5$$

The cumulative frequency (c.f.), equal or just greater than  $\frac{N+1}{2}$  is 65 and value corresponding to 65 is 5.

$\therefore$  ***Median is 5.***

## (ii) Continuous Frequency Distribution:

In case of continuous frequency distribution, the class corresponding to the c.f. just greater than  $\frac{N}{2}$  is called the ***median class*** and the value of the median is obtained using the formula:

$$\text{Median} = l + \frac{h}{f} \left( \frac{N}{2} - c \right)$$

$l$  - is the lower limit of the median class

$f$  - is the frequency of the median class

$h$  - is the magnitude of the median class

$c$  - is the c.f. of the class preceding the median class

Ex: Find the median of the following data:

| <b>Wages<br/>(in Rs.)</b> | 2000-3000 | 3000-4000 | 4000-5000 | 5000-6000 | 6000-7000 |
|---------------------------|-----------|-----------|-----------|-----------|-----------|
| <b>No. of<br/>workers</b> | 3         | 5         | 20        | 10        | 5         |

*Solution:*

| <i><b>Wages<br/>(in Rs.)</b></i> | <i><b>No. of<br/>Employees</b></i> | <i><b>c.f.</b></i> |
|----------------------------------|------------------------------------|--------------------|
| 2000-3000                        | 3                                  | 3                  |
| 3000-4000                        | 5                                  | 8                  |
| 4000-5000                        | 20                                 | 28                 |
| 5000-6000                        | 10                                 | 38                 |
| 6000-7000                        | 5                                  | 43                 |
|                                  | N= 43                              |                    |

$$\frac{N}{2} = \frac{43}{2} = 21.5$$

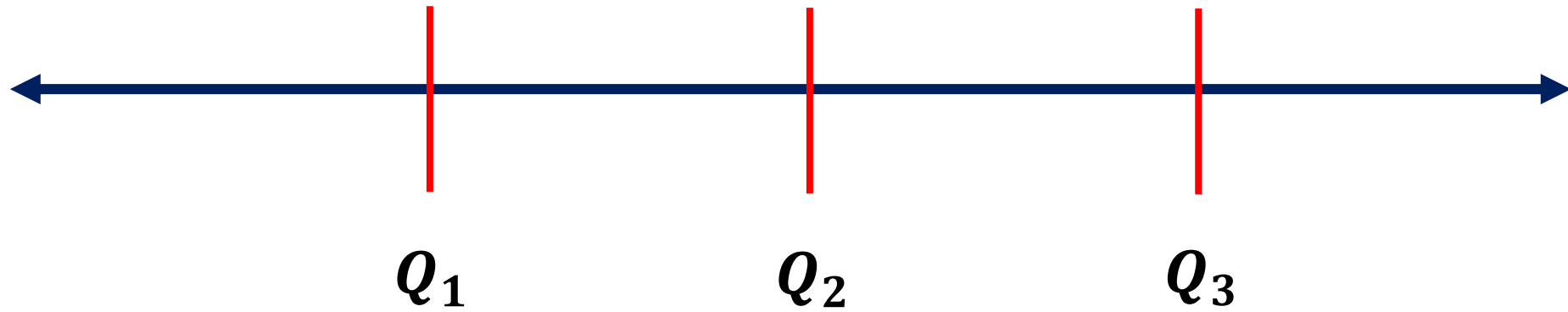
Cumulative frequency just greater than 21.5 is 28 and the corresponding class is 4000-5000

Thus the median class is 4000-5000

$$l = 4000; h = 1000; f = 20; c = 8$$

$$\text{Median} = 4000 + \frac{1000}{20} (21.5 - 8) \Rightarrow \therefore \text{Median} = 4675.$$





# Quartiles

First Quartile ( $Q_1$ ) = Size of  $\frac{N+1}{4}$  *th* item (Discrete series)

{ See the cumulative frequency (c.f.) value, equal or just greater than the  $\frac{N+1}{4}$  }

Continuous series:

$Q_1$  = Size of  $\frac{N}{4}$  *th* item

$$Q_1 = l + \frac{\frac{N}{4} - c.f.}{f} \times i$$

Third Quartile  $Q_3$  = Size of  $3 \left( \frac{N+1}{4} \right)$  *th* item (Discrete series)

{See the cumulative frequency (c.f.) value, equal or just greater than the  $3 \left( \frac{N+1}{4} \right)$ }

Continuous series:

$Q_3$  = Size of  $\frac{3N}{4}$  *th* item.

$$Q_3 = l + \frac{\frac{3N}{4} - c.f.}{f} \times i$$

**Example 3:** Calculate  $Q_1$  and  $Q_3$  for the following data.

| Roll No. | 1  | 2  | 3  | 4  | 5  | 6  | 7  |
|----------|----|----|----|----|----|----|----|
| Marks    | 20 | 28 | 40 | 12 | 30 | 15 | 50 |

**Solution:** Marks in ascending order    12 15 20 28 30 40 50

$$Q_1 = \text{Size of } \frac{N+1}{4} \text{ th item} = \text{Size of } \frac{7+1}{4} = 2^{\text{nd}} \text{ item.}$$

Size of 2<sup>nd</sup> item is 15.    Hence  $Q_1 = 15$

$$Q_3 = \text{Size of } 3\left(\frac{N+1}{4}\right) \text{ th item} = \text{Size of } 3\left(\frac{7+1}{4}\right) = 6^{\text{th}} \text{ item.}$$

Size of 6<sup>th</sup> item is 40.    Hence  $Q_3 = 40$

**Example 4:** Compute the value of  $Q_1$  and  $Q_3$  for following data:

| <i>C.I.</i> | 10-20 | 20-30 | 30-40 | 40-50 | 50-60 | 60-70 | 70-80 |
|-------------|-------|-------|-------|-------|-------|-------|-------|
| <i>f</i>    | 12    | 19    | 5     | 10    | 9     | 6     | 6     |

**Solution:**

| <i>Marks</i> | <i>Frequency</i> | <i>Cumulative<br/>Frequency</i> |
|--------------|------------------|---------------------------------|
| 10-20        | 12               | 12                              |
| 20-30        | 19               | 31                              |
| 30-40        | 5                | 36                              |
| 40-50        | 10               | 46                              |
| 50-60        | 9                | 55                              |
| 60-70        | 6                | 61                              |
| 70-80        | 6                | 67                              |
|              | <i>N</i> = 67    |                                 |

$$Q_1 = \text{Size of } \frac{N}{4} \text{ th item} = \text{Size of } \frac{67}{4} = 16.75^{\text{th}} \text{ item.}$$

$Q_1$  - lies in the interval **20-30**

$$Q_1 = l + \frac{\frac{N}{4} - c.f.}{f} \times i$$

$$l = 20, N/4 = 16.75, c.f. = 12$$

$$f = 19, i = 10$$

$$Q_1 = 20 + \frac{\frac{67}{4} - 12}{19} \times 10 = 20 + 2.5 = 22.5$$

Hence  $Q_1 = 22.5$

$$Q_3 = \text{Size of } \frac{3N}{4} \text{ th item} = \text{Size of } \frac{3 \times 67}{4} = 50.25^{\text{th}} \text{ item.}$$

$Q_3$  - lies in the class **50-60**.

$$Q_3 = l + \frac{\frac{3N}{4} - c.f.}{f} \times i$$

$l = 50, 3N/4 = 50.25, c.f. = 46$   
 $f = 9, i = 10$

$$Q_3 = 50 + \frac{50.25 - 46}{9} \times 10 = 50 + 4.72 = 54.72$$

Hence  $Q_3 = 54.72$



**Deciles:**  $D_4$  = Size of  $4 \left( \frac{N+1}{10} \right)$  item in individual and discrete series.

$D_4$  = Size of  $\frac{4N}{10}$  th item in continuous series.

**Percentiles:**  $P_{60}$  = Size of  $60 \left( \frac{N+1}{100} \right)$  th item in discrete series.

$P_{60}$  = Size of  $\frac{60N}{100}$  th item in continuous series.

$D_4$  = Size of  $\frac{4N}{10}$  th item =  $\frac{4 \times 67}{10} = 26.8$  th item.

$D_4$  lies in the interval of 20-30.

$$D_4 = l + \frac{\frac{4N}{10} - c.f.}{f} \times i$$

$$l = 20, 4N/10 = 26.8, c.f. = 12$$

$$f = 19, i = 10$$

$$D_4 = 27.79$$

$P_{60}$  = Size of  $\frac{60N}{100}$  th item =  $\frac{60 \times 67}{100} = 40.2$  th item

$P_{60}$  lies in the interval of 40-50.

$$P_{60} = l + \frac{\frac{60N}{100} - c.f.}{f} \times i$$

$$l = 40, 60N/100 = 40.2, c.f. = 36$$

$$f = 10, i = 10$$

$$P_{60} = 44.2$$

# MODE

*Definition:* Mode is the value of the variable which is predominant in the series.

1. For a discrete frequency distribution, mode is the value of  $x$  – corresponding to maximum frequency.

|     |   |   |    |    |    |    |   |   |
|-----|---|---|----|----|----|----|---|---|
| $x$ | 1 | 2 | 3  | 4  | 5  | 6  | 7 | 8 |
| $f$ | 4 | 9 | 16 | 25 | 22 | 15 | 7 | 3 |

# Continuous Frequency distribution:

$$\text{Mode} = l + \frac{h(f_1 - f_0)}{(f_1 - f_0) - (f_2 - f_1)} = l + \frac{h(f_1 - f_0)}{2f_1 - f_0 - f_2}$$

$l$  – lower limit of the modal class

$h$  – magnitude of the modal class

$f_1$  - frequency of the modal class

$f_0$  and  $f_2$  - frequencies of the class preceding and succeeding the modal class

| C.I.      | 0-10 | 10-20 | 20-30 | 30-40    | 40-50    | 50-60    | 60-70 | 70-80 |
|-----------|------|-------|-------|----------|----------|----------|-------|-------|
| Frequency | 5    | 8     | 7     | 12 $f_0$ | 28 $f_1$ | 20 $f_2$ | 10    | 10    |

Hence the maximum frequency is 28

Thus the modal class is 40-50

$$\text{Mode} = 40 + \frac{10(28-12)}{2 \times 28 - 12 - 20} = 46.67$$

A distribution is having only one mode is called *Unimodal*.

If it contains more than one mode, it is called *bimodal* or *multimodal*.

Then the value of the mode cannot be determined the above formula and hence mode is *ill-defined*.

$$\textbf{Mode} = 3 \textbf{ Median} - 2 \textbf{ Mean}$$

1) Calculate the mean, median and mode of the following data.

| <b><i>Wages<br/>(in lakhs.)</i></b> | 15-19 | 20-24 | 25-29 | 30-34 | 35-39 | 40-44 | 45-49 | 50-54 | 55-59 | 60-64 |
|-------------------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| <b><i>No. of<br/>workers</i></b>    | 31    | 47    | 59    | 78    | 104   | 113   | 81    | 60    | 52    | 25    |

Mean = 39.52

Median = 39.77

Mode = 40.6



2)

| <i>Variable</i>  | 10-13 | 13-16 | 16-19 | 19-22 | 22-25 | 25-28 | 28-31 | 31-34 | 34-37 | 37-40 |
|------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| <i>Frequency</i> | 8     | 15    | 27    | 51    | 75    | 54    | 36    | 18    | 9     | 7     |

Calculate the Mean, Median and Mode for the following data.

Mean = 24.19

Median = 23.96

Mode = 23.6

**The following distribution gives the pattern of overtime work per week done by 100 employees of a company.**  
**Calculate  $Q_1$ ,  $D_7$  and  $P_{60}$ .**

| <b>Overtime<br/>(hours)</b> | 10-15 | 15-20 | 20-25 | 25-30 | 30-35 | 35-40 |
|-----------------------------|-------|-------|-------|-------|-------|-------|
| <b>No. of<br/>Employees</b> | 11    | 20    | 35    | 20    | 8     | 6     |

$$\begin{aligned}\text{Med} &= l + \frac{(n/2) - cf}{f} \times h \\ &= 20 + \frac{50 - 31}{35} \times 5 = 20 + 2.714 = 22.714 \text{ hours}\end{aligned}$$

$Q_1$  = value of  $(n/4)$  th observation = value of  $(100/4)$ th = 25th observation

$$\text{Thus } Q_1 = l + \frac{(n/4) - cf}{f} \times h = 15 + \frac{25 - 11}{20} \times 5 = 15 + 3.5 = 18.5 \text{ hours}$$

$D_7$  = value of  $(7n/10)$ th observation = value of  $(7 \times 100)/10 = 70$ th observation

$$\text{Thus } D_7 = l + \frac{(7n/10) - cf}{f} \times h = 25 + \frac{70 - 66}{20} \times 5 = 25 + 1 = 26 \text{ hours}$$

$P_{60}$  = Value of  $(60n/100)$  th observation =  $60 \times (100/100) = 60$ th observation

$$\text{Thus } P_{60} = l + \frac{(60 \times n/100) - cf}{f} \times h = 20 + \frac{60 - 31}{35} \times 5 = 24.14 \text{ hours}$$

| <i>Mutual Funds<br/>Price (Rs.)</i> | <i>Upper Class<br/>Boundary</i> | <i>Number of Funds<br/>(f)</i> | <i>Cumulative Frequency</i> |                  |
|-------------------------------------|---------------------------------|--------------------------------|-----------------------------|------------------|
|                                     |                                 |                                | <i>Less than</i>            | <i>More than</i> |
| 10–15                               | 15                              | 6                              | 6                           | 40               |
| 15–20                               | 20                              | 11                             | $6 + 11 = 17$               | $40 - 6 = 34$    |
| 20–25                               | 25                              | 9                              | $17 + 9 = 26$               | $34 - 11 = 23$   |
| 25–30                               | 30                              | 7                              | $26 + 7 = 33$               | $23 - 9 = 14$    |
| 30–35                               | 35                              | 5                              | $33 + 5 = 38$               | $14 - 7 = 7$     |
| 35–40                               | 40                              | 2                              | $38 + 2 = 40$               | $7 - 5 = 2$      |

